Lab Exercise 7

Setting up Jenkins Pipeline Job from Git

Objective: To set up a Jenkins pipeline job from Git version control system to enable automated CI for building, testing, and potentially deploying software upon code changes

Tools required: Jenkins, Git and Linux

Prerequisites: None

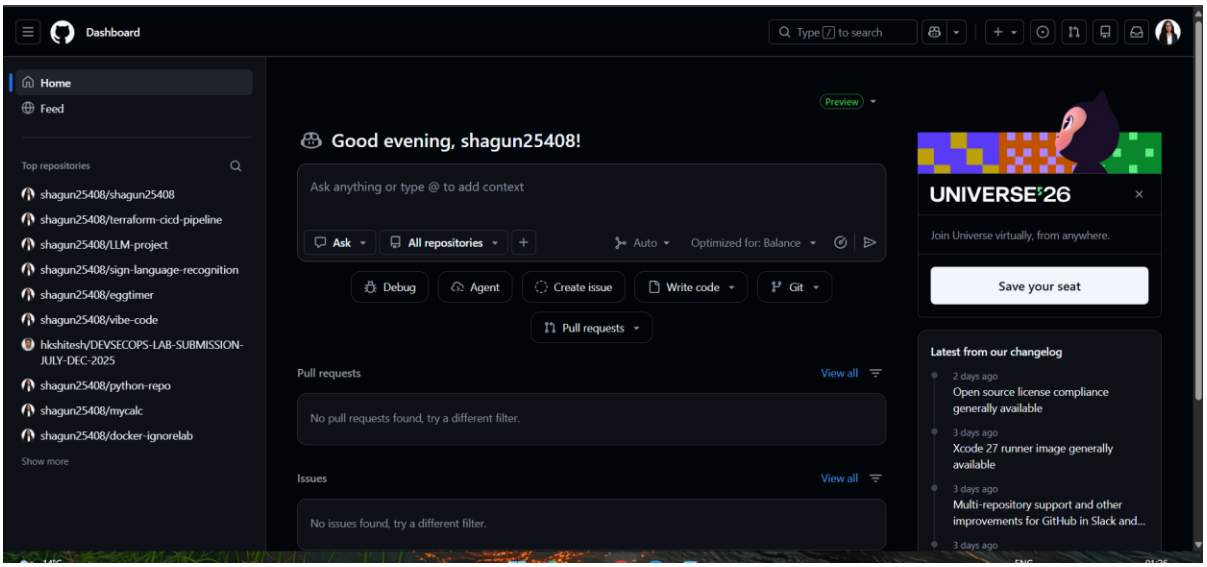
Steps to be followed:

1. Create a Git repository
2. Push the pipeline script into the Git repository
3. Create a pipeline script-based freestyle job

Step 1: Create a Git repository

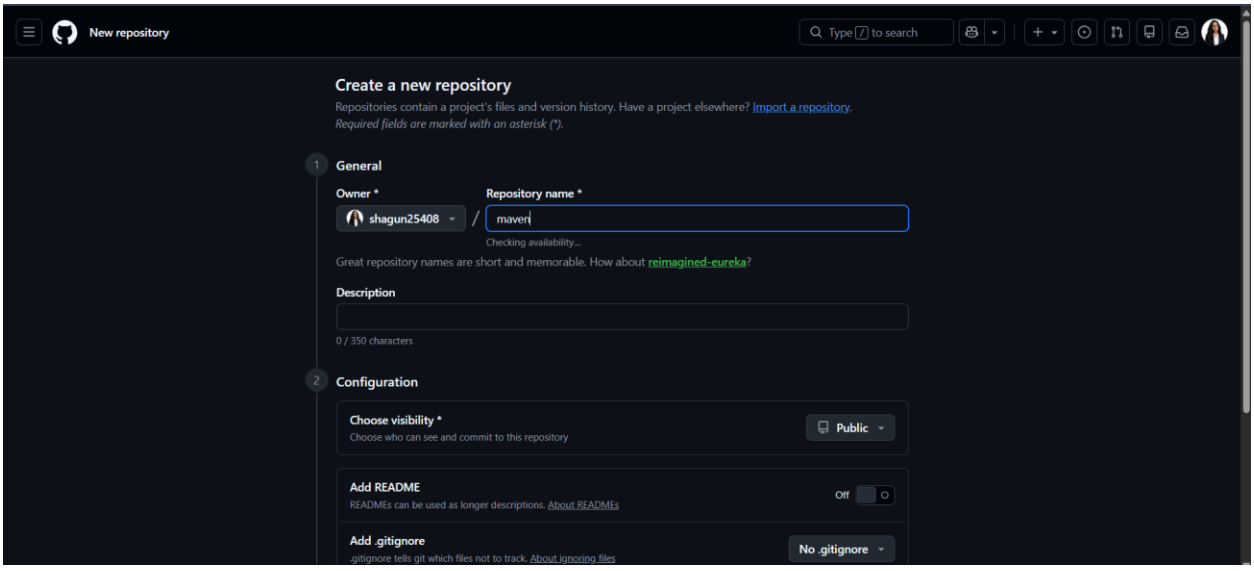
1.1 Open the browser in your lab, go to **github.com**, and **Sign in** to your account

Note: If you do not have a GitHub account, visit the official website at **https://github.com/signup** and create a new account

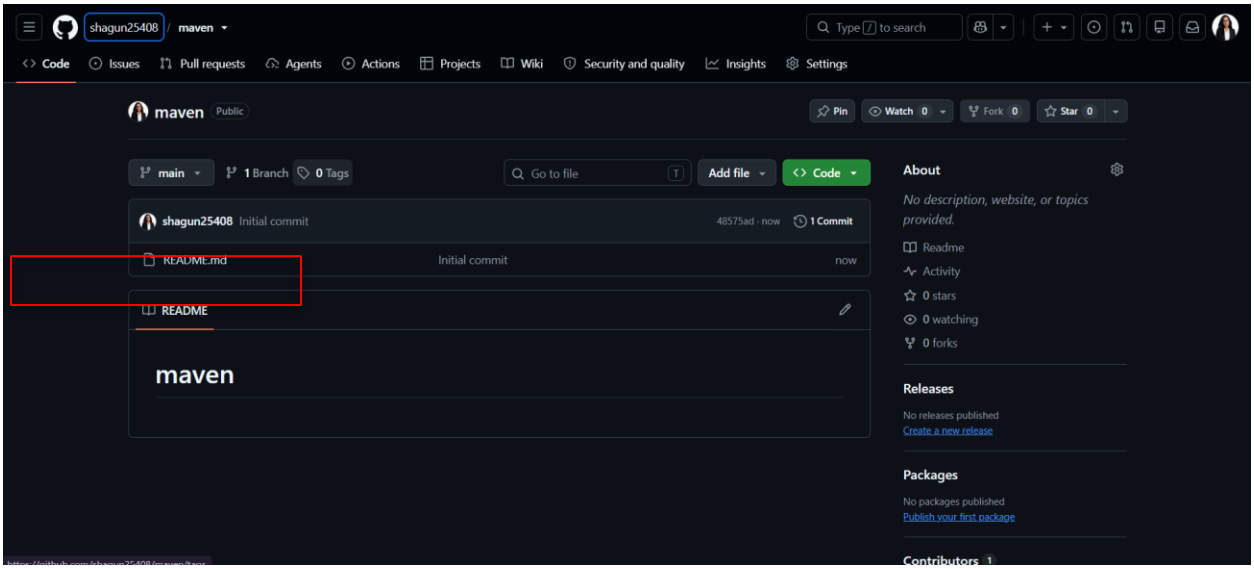


1.2 Click on **New** as shown in the screenshot below:

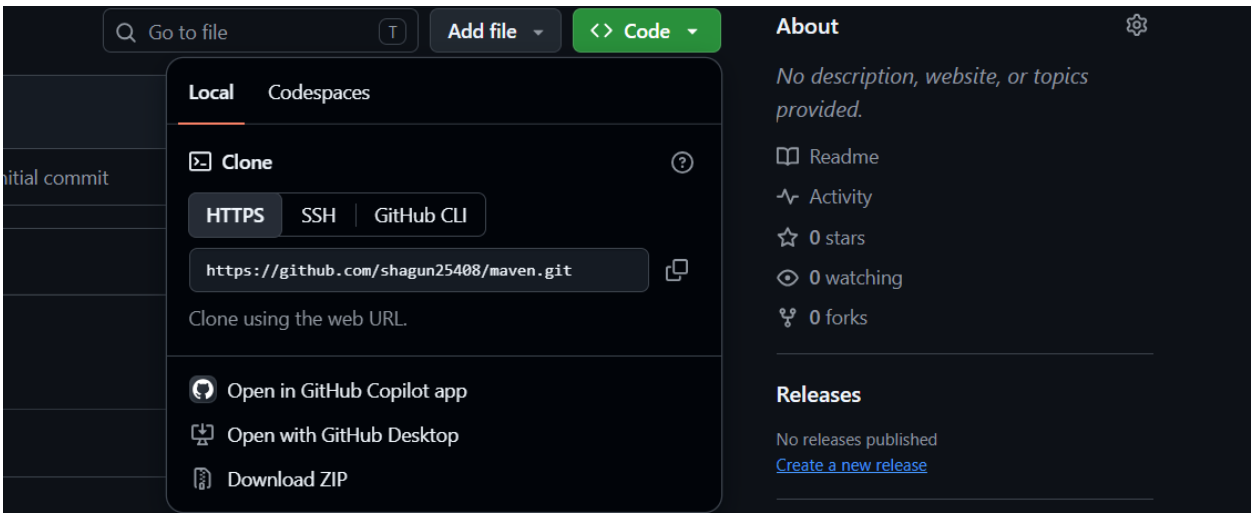
1.3 Enter a desired name for your repository and choose **Public** as shown in the screenshot below:



1.4 Click on **Add a README file** and then click on **Create repository** as shown in the screenshot below:



1.5 Click on **Code** as shown in the screenshot below:

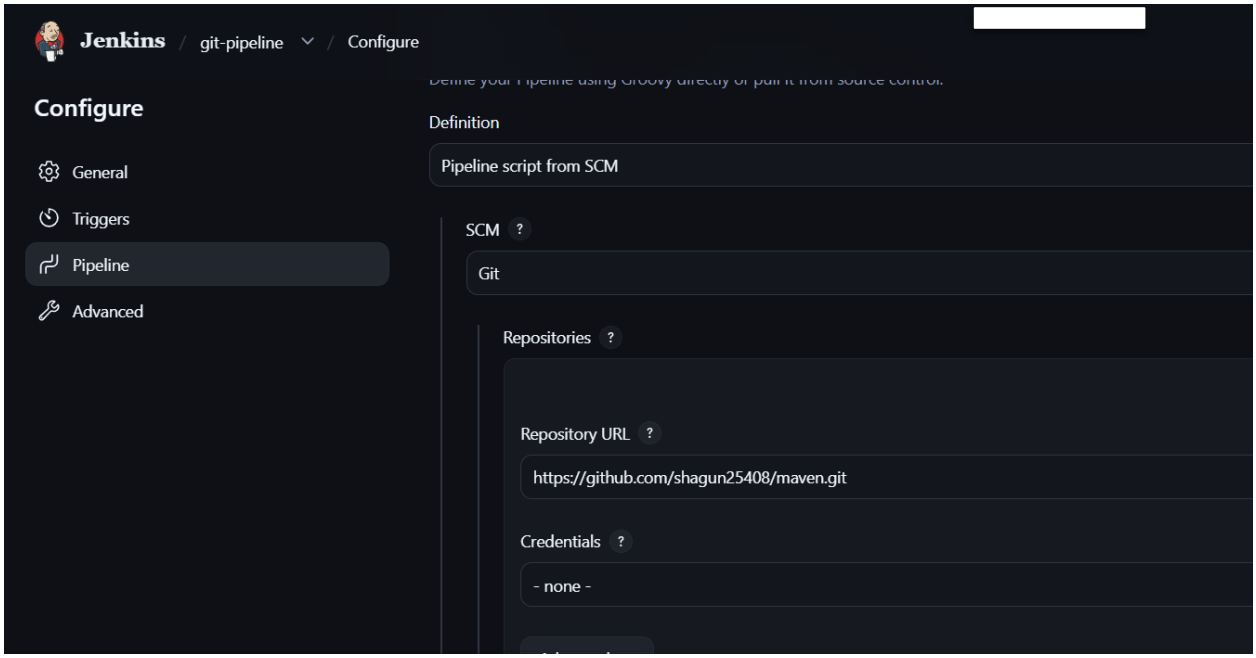


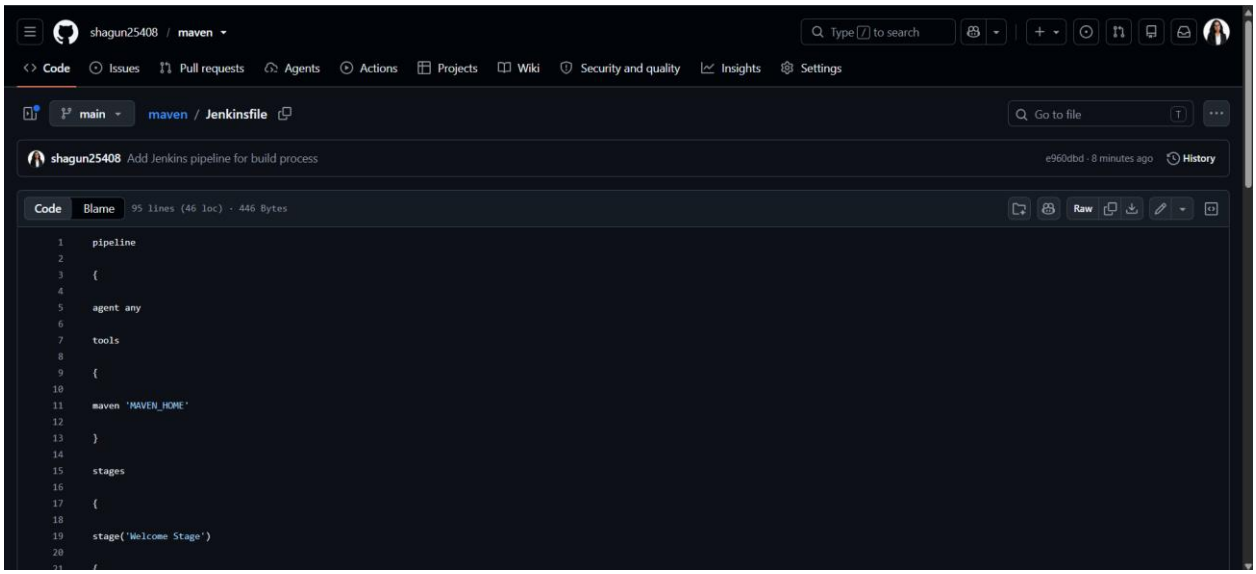
1.6 Copy the repository URL as shown in the screenshot below:

https://github.com/shagun25408/maven.git

Step 2: Push the pipeline script into the Git repository

2.1 Open the Linux terminal in your lab and clone the repository using the below command:
git clone RepositoryURL





2.2 Paste the below pipeline script inside the file as shown in the screenshot below:

```
pipeline
{
agent any
        tools
```

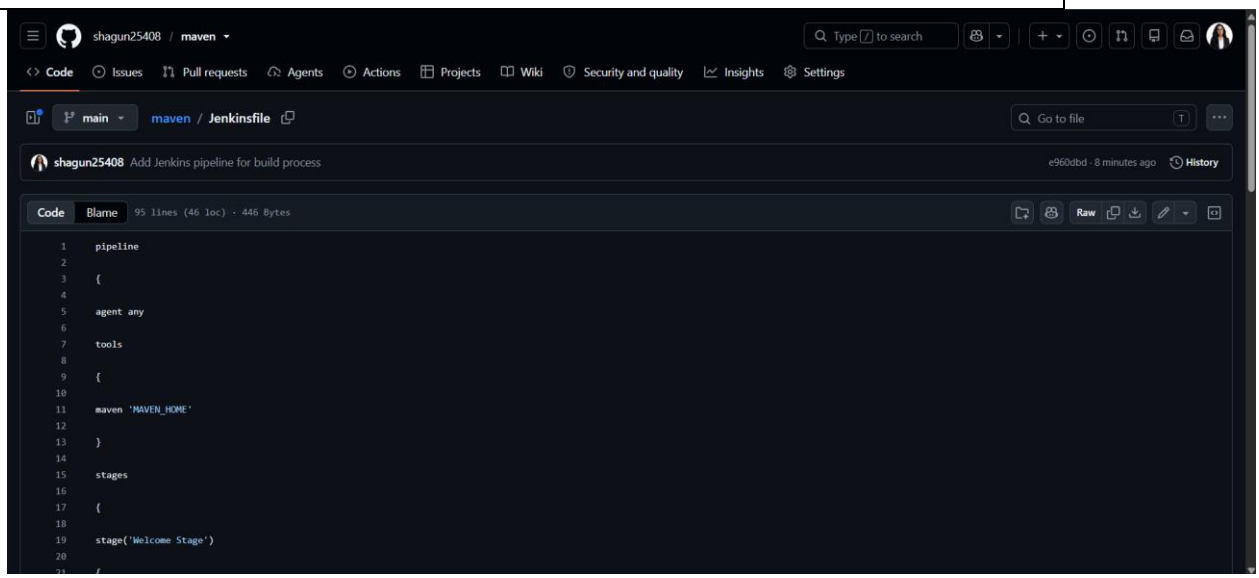
```
{
        maven 'MAVEN_HOME'
}
stages
{
        stage('Welcome Stage')
        {
                steps
                {
                        echo 'Welcome to Pipeline'
                }
        }

        stage('Clean Stage')
        {
                steps
                {
                        bat 'mvn clean'
                }
        }
        stage('Build Stage')
        {
                steps
                {
                        bat 'mvn install'
                }
        }
        stage('Build Success')
        {
                steps
                {
                        echo 'Build Success'
                }
        }

        stage('Final Success')
        {
                steps
                {
                        echo 'Final Success'
                }
        }
```

```
        }
}
```

Note: Ensure you provide your Git repository URL on line7, save, and exit the page by clicking on **ctrl+S** to save and **ctrl+X** to exit



2.3 Stage and commit the changes using the below commands:

git add .

git commit -m "initial commit"

```
syedsharozsimpl@ip-172-31-40-171:~/Maven$ git add .
syedsharozsimpl@ip-172-31-40-171:~/Maven$ git commit -m "initial commit"
[master a0e2b3a] initial commit
 1 file changed, 41 insertions(+)
 create mode 100644 demofile
syedsharozsimpl@ip-172-31-40-171:~/Maven$
```

2.4 Push the file to the Git repository using the below command:

git push

```
syedsharozsimpl@ip-172-31-40-171:~/Maven$ git push
Username for 'https://github.com': Poojahk10
Password for 'https://Poojahk10@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 614 bytes | 614.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Poojahk10/Maven.git
   0d9b8d3..a0e2b3a  master -> master
syedsharozsimpl@ip-172-31-40-171:~/Maven$
```

2.5 Navigate to your Git repository to check for the file that is pushed as shown in the screenshot below:

